COMPETITIVE ANALYSIS

COMPARATIVE ANALYSIS.

75% LESS CAPEX. 45% LESS OPEX.

A detailed comparison of Tobe Energy's electrolyzer against Alkaline, PEM, Solid Oxide, and Anion Exchange systems across cost, efficiency, and longevity.

>92%	8×	<30°C	0
SYSTEM EFFICIENCY	LESS HEAT LOSS	OPERATING TEMP	RARE EARTH METALS

© 2026 TOBE ENERGY INC.

COMPLETE LIFECYCLE GHG EMISSIONS ANALYSIS FOR TOBE ENERGY HYDROGEN SYSTEM

Following DOE 45VH2-GREET Methodology (Rev. December 2025) **Target Year: 2026 | Location: Oklahoma (Plains NEEDS Region)**

---

1. ELECTRICITY FOR ELECTROLYSIS

Stack electricity consumption: 42 kWh/kg H_2 (HHV basis) **Electricity source:** 100% renewable PPA (wind, Oklahoma)

Critical Assumption & Emission Factor Selection

- The GREET 2025 extract shows wind as "0 gCO_2eq/kWh" for operational emissions only.
- Per 45VH2-GREET methodology (Section 3.2.1), renewable electricity must include full lifecycle emissions (manufacturing, construction, decommissioning).
- GREET 2025 default for onshore wind (2026): **14.0 gCO_2eq/kWh** (conservative value; GREET v2020 = 13.1, but 2025 update accounts for supply chain emissions. Source: GREET 2025 Workbook, "Electricity Generation" tab, "Wind - Onshore" pathway).
- Why conservative? Industry data shows 11–14 gCO_2eq/kWh for 2026 wind (NREL 2023). Using 14.0 gCO_2eq/kWh aligns with DOE's "overestimate where uncertain" directive.

Calculation = 42 kWh/kg × 14.0 gCO_2eq/kWh ÷ 1,000 = **0.588 $kgCO_2eq/kg\ H_2$**

---

2. COMPRESSION ENERGY (1 bar → 200 bar)

System: Reciprocating multi-stage compressor with intercooling (200 bar output)

Compression Work Calculation (First Principles)

- Pressure ratio: $r = \frac{P_{out}}{P_{in}} = \frac{200}{1} = 200$
- Optimal staging: For reciprocating compressors with intercooling, use polytropic compression (n=1.2) and 3 stages (industry standard for 200 bar).
  - Stage pressure ratio: $r = 200^{1/3} = 5.85$
- Work per stage (polytropic):

$$W_{stage} = \frac{n}{n-1} \cdot R \cdot T_{in} \cdot \eta_{comp}^{-1} \cdot \left(r^{\frac{n-1}{n}} - 1\right)$$

Where:

- $R = 8.314$ J/mol·K, $M_{H_2} = 2.016$ g/mol → $R_{spec} = 4.124$ kJ/kg·K
- $T_{in} = 298$ K (standard inlet temp)
- $\eta_{comp} = 0.85$ (conservative compressor efficiency, NREL 2022)
- $r^{\frac{n-1}{n}} = 5.85^{\frac{0.2}{1.2}} = 5.85^{0.1667}$

© 2026 TOBE ENERGY INC.

$$W_{stage} = \frac{1.2}{0.2} \cdot 4.124 \cdot 298 \cdot 0.85^{-1} \cdot (5.85^{0.1667} - 1) = 6{,}229 \text{ kJ/kg} = 1.73 \text{ kWh/kg H}_2$$

- Total work (3 stages): $W_{total} = 3 \times 1.73 = 5.19$ kWh/kg H_2
- Add motor & electrical losses (conservative):
  - Motor efficiency: 95% → $\frac{5.19}{0.95} = 5.46$ kWh/kg H_2
  - Electrical losses (transformers, wiring): +5% → $5.46 \times 1.05 = 5.73$ kWh/kg H_2

Validation: NREL 2022 reports 2.5–3.5 kWh/kg for 200-bar compression.

Emissions Calculation

- Electricity source: 100% renewable PPA (same as electrolysis).

= 5.73 kWh/kg × 14.0 gCO_2eq/kWh ÷ 1,000 = **0.0802 $kgCO_2eq$/kg H_2**

---

3. WATER TREATMENT (RO SYSTEM)

System: Small RO unit with 3–5 HP pump (use 5 HP for conservatism)

Energy Consumption Calculation

- Pump power: $5 \text{ HP} = 3.73$ kW
- Water consumption: 10 L/kg H_2 (high-purity electrolysis, conservative end of 9–10 L/kg range)
- RO recovery rate: 75% (industry standard) → Feedwater required = $\frac{10}{0.75} = 13.33$ L H_2O/kg H_2
- Annual production: 900,000 kg H_2/year (nameplate)
- Annual feedwater: $900{,}000 \text{ kg H}_2 \times 13.33 \text{ L/kg} = 12{,}000{,}000$ L
- RO runtime: Assumes continuous operation with electrolyzer (8,760 hours/year at 100% capacity factor).
- Annual RO energy: $3.73 \text{ kW} \times 8{,}760 = 32{,}675$ kWh
- Energy per kg H_2:

$$\frac{32{,}675 \text{ kWh}}{900{,}000 \text{ kg H}_2} = 0.0363 \text{ kWh/kg H}_2$$

Emissions Calculation

- Electricity source: 100% renewable PPA.

= 0.0363 kWh/kg × 14.0 gCO_2eq/kWh ÷ 1,000 = **0.0005 $kgCO_2eq/kg\ H_2$**

---

4. AUXILIARY LOADS

Components: Control systems (negligible), winter heating (marginal contribution)

Energy Consumption Calculation

- Winter heating: 2 kW average (conservative), Oklahoma heating season = 180 days → $2 \text{ kW} \times 24 \text{ hrs/day} \times 180 = 8{,}640$ kWh.
  - Conservative adjustment: Assume 50% derating → $2 \text{ kW} \times 24 \text{ hrs/day} \times 180 \times 0.5 = 4{,}320$ kWh.
- Energy per kg H_2:

$$\frac{4{,}320 \text{ kWh}}{900{,}000 \text{ kg H}_2} = 0.0048 \text{ kWh/kg H}_2$$

- Control systems: 5 Arduino boards (total <0.1 kW) → negligible (<0.0001 kWh/kg H_2).

Emissions Calculation

- Electricity source: 100% renewable PPA.

= 0.0048 kWh/kg × 14.0 gCO_2eq/kWh ÷ 1,000 = **0.0001 $kgCO_2eq/kg\ H_2$**

---

5. WATER SUPPLY CHAIN EMISSIONS

Source: Municipal water supply (per GREET methodology)

Emission Factor & Calculation

- GREET 2025 default for municipal water: 110 gCO_2eq/m^3 (conservative; Source: GREET 2025 Workbook, "Water Supply" tab).
- Water consumption: 10 L/kg H_2 = 0.010 m^3/kg H_2.

= 0.010 m^3/kg × 110 gCO_2eq/m^3 ÷ 1,000 = **0.0011 $kgCO_2eq/kg\ H_2$**

---

6. TOTAL LIFECYCLE GHG EMISSIONS (WELL-TO-GATE)

Summary of Emissions Contributors

COMPONENT	ENERGY (KWH/KG H_2)	EMISSION FACTOR (GCO_2EQ/KWH)	EMISSIONS ($KGCO_2EQ$/KG H_2)
Electrolysis stack	42.000	14.0	0.5880
Compression	5.730	14.0	0.0802
Water treatment (RO)	0.0363	14.0	0.0005
Auxiliary loads	0.0048	14.0	0.0001
Water supply chain	—	110 gCO_2eq/m^3	0.0011
TOTAL	**47.771**	—	**0.6699**

© 2026 TOBE ENERGY INC.

Final Calculation = 0.5880 + 0.0802 + 0.0005 + 0.0001 + 0.0011 = **0.6699 $kgCO_2eq/kg\ H_2$**

SENSITIVITY ANALYSIS

Key Variables & Impact on Total Emissions

VARIABLE	BASE CASE (CONSERVATIVE)	PESSIMISTIC	OPTIMISTIC	IMPACT ON TOTAL ($KGCO_2EQ$/KG H_2)
Wind PPA emission factor	14.0	16.0 (older turbines)	11.0 (advanced turbines)	0.719 → 0.562
Compression energy	5.73	6.70 (low efficiency)	4.75 (high efficiency)	0.681 → 0.658
Water consumption	10 L/kg	12 L/kg	9 L/kg	0.671 → 0.669
Total range	—	—	—	**0.681 → 0.562**

- Critical insight: Wind PPA emission factor dominates (85% of total). Even with optimistic assumptions (11.0 gCO_2eq/kWh + 4.75 kWh/kg compression), total = 0.562 $kgCO_2eq/kg\ H_2$ > 0.45 threshold.

Why Wind PPA ≠ 0 Emissions? GREET 2025 includes:

- Turbine manufacturing (steel, concrete, rare earths): 8.2 gCO_2eq/kWh
- Construction & installation: 3.1 gCO_2eq/kWh
- O&M & decommissioning: 2.7 gCO_2eq/kWh
- Source: GREET 2025 Workbook, "Wind - Onshore" pathway (2026 projection).

COMPARISON TO 45V CREDIT TIERS

© 2026 TOBE ENERGY INC.

LIFECYCLE GHG ($KGCO_2EQ/KG\ H_2$)	45V CREDIT TIER	TOBE ENERGY RESULT
< 0.45	$3.00/kg	—
0.45 – 1.5	$1.00/kg	✓ QUALIFIES HERE
1.5 – 2.5	$0.75/kg	—
2.5 – 4.0	$0.60/kg	—

- Threshold analysis:
  - Maximum allowable for $3.00/kg tier: **0.45 $kgCO_2eq/kg\ H_2$**.
  - Tobe's emissions (0.6699) exceed this by **48.9%**.
  - Even with optimistic assumptions (0.562), emissions remain 24.9% above threshold.

CONCLUSION & RECOMMENDATIONS

1. Total lifecycle GHG emissions: **0.670 $kgCO_2eq/kg\ H_2$** (well-to-gate, conservative estimate).
2. 45V qualification:
   - Does NOT qualify for $3.00/kg tier (emissions > 0.45 $kgCO_2eq/kg\ H_2$).
   - Qualifies for **$1.00/kg tier** (emissions < 1.5 $kgCO_2eq/kg\ H_2$).
3. Financial impact vs. preliminary analysis:
   - Preliminary analysis incorrectly assumed wind PPA = 0 gCO_2eq/kWh (operational only).
   - Actual annual PTC revenue: **$900,000/facility** (vs. claimed $2.7M).
   - Revised 10-year net benefit (PPA @ $0.030/kWh):
     - PPA cost: $1.13M/year
     - PTC revenue: $0.9M/year
     - Net loss: **-$230,000/year** (vs. claimed +$1.57M/year).

Critical Path Forward

- To qualify for $3.00/kg tier: Reduce emissions by 32.8% (to ≤0.45 kgCO_2eq/kg H_2).
  - Option 1: Use wind PPA with certified low-impact manufacturing (e.g., <10.7 gCO_2eq/kWh).
  - Option 2: Add carbon-free compression (e.g., waste-heat recovery to cut compression energy by 30%).
- Immediate action: Re-run GREET model with facility-specific PPA documentation (e.g., RECs with embodied carbon data).

© 2026 TOBE ENERGY INC.

Final determination: Tobe Energy's system does not qualify for the $3.00/kg 45V credit tier under current specifications and GREET 2025 methodology. Rigorous accounting of renewable lifecycle emissions is non-negotiable for 45V compliance.

Prepared by: Colby DeWeese, CEO — Tobe Energy Corp **Date:** April 10, 2026 **Methodology Compliance:** DOE 45VH2-GREET (Rev. December 2025), Section 4.3 ("Full Lifecycle Boundaries for Renewable Electricity")

TOBE ENERGY CORP — CONFIDENTIAL